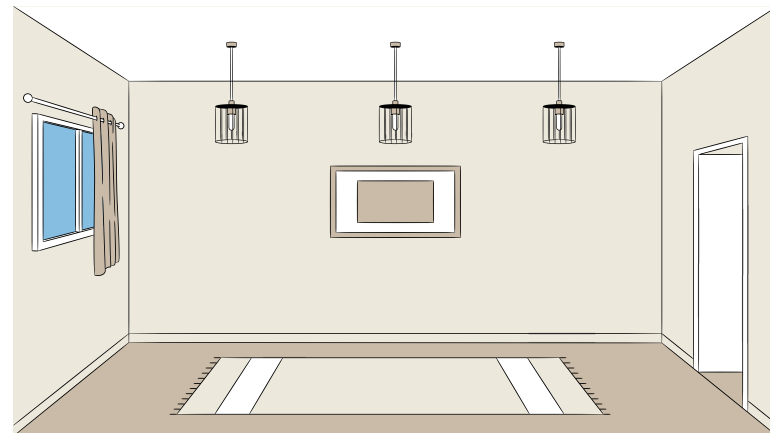


The 3 Musketeers of Web Dev

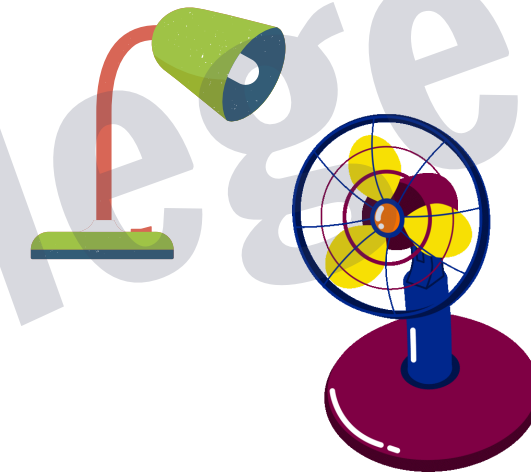
HTML
(structure)



CSS
(style)



JS
(logic)



Starter Code

<style> tag connects HTML with CSS

<script> tag connects HTML with JS

Apna College

```
<html>
```

```
  <head>
```

```
    <title> Website Name </title>
```

```
  </head>
```

```
  <body>
```

```
    <!-- Content Tags -->
```

```
  </body>
```

```
</html>
```

you can write js code like prompt or loops strings or alert etc inside the script tag only , if u dont want to make a new file and link it to the html file

like

```
<body>  
<script>  
alert("hello");  
</script>  
</body>  
</html>
```

Window Object

The window object represents an open window in a browser. It is browser's object (not JavaScript's) & is automatically created by browser.

(can be asked in interviews so remember these 2 lines)

It is a **global** object with lots of properties & methods.

it is a browsers object (the window is a object name present in browser or in any web page)
go type window in the console ,,
itll show a object with alot of functions in it

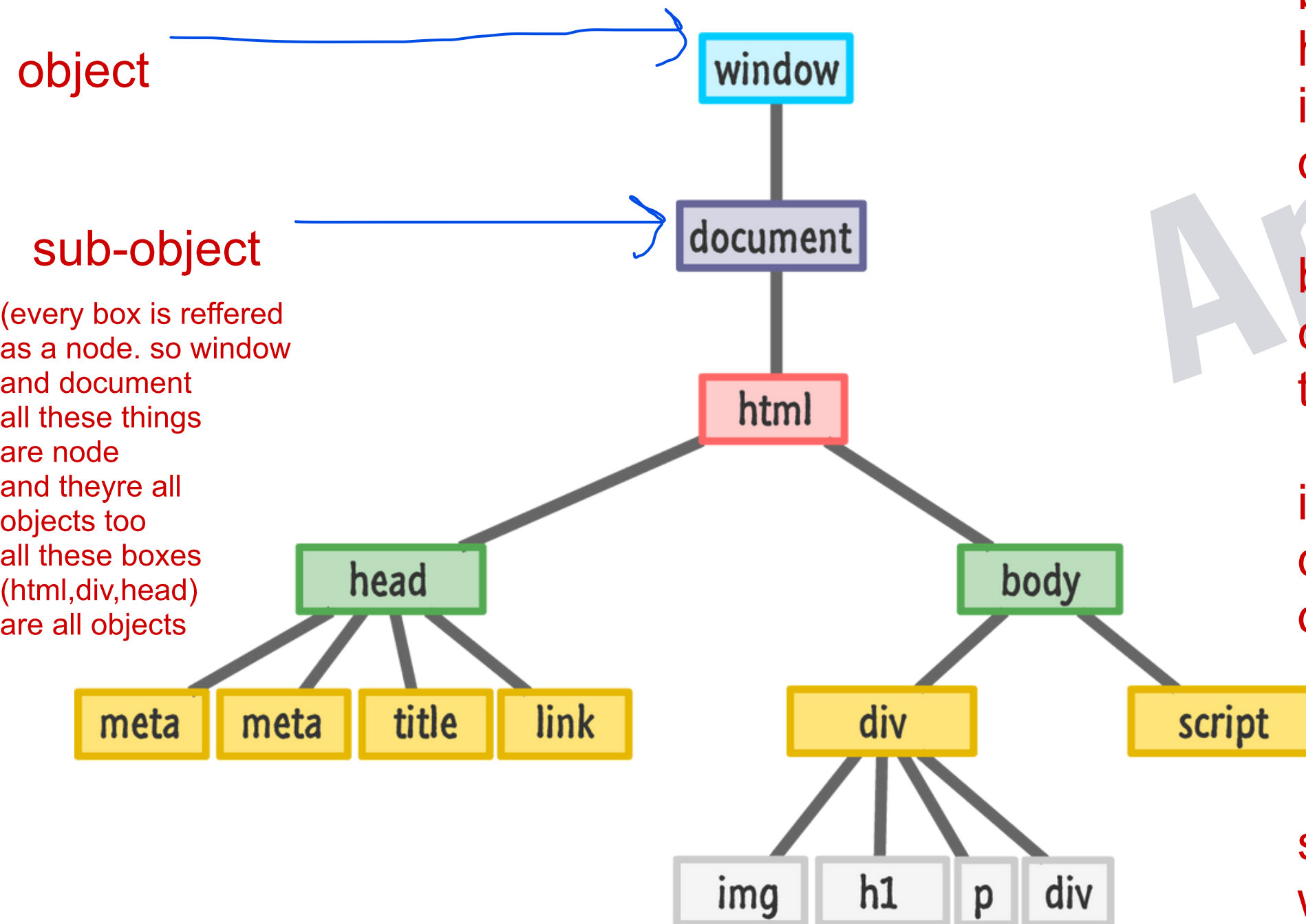
"by browser" so if using chrome , chrome makes it and if u using edge edge makes it (makes window object);

every code you write in script.js or any js code ,, all that code comes under window object which is automatically created , so all the js code you right is inside and is a part of window object see code line 1 and 2 theyre the same. you dont need to write window. coz the code automatically understands whatever youre writing is inside the window object only.

this tree like structure is exactly what the DOM is , document object model is a tree like structure shown below

What is DOM?

When a web page is loaded, the browser creates a **Document Object Model (DOM)** of the page



basically we all know that html css js are all part of the html code only.....
if you want to get the html code from the js file you can do it...

because as yk window is the largest object it has all the code inside it html code js code .. all the code is inside the window object but where in it?

inside the window object, there's another object named document , so document is an object inside the window object which stores the html code, only html code it stores

so the Document Object Model is:
window-->document-->html code

there's also a code as `console.dir` unlike `console.log` which is used to print all the properties and methods of some special objects like here for ex document you can do `console.dir(window.document)` to see the properties and methods of the document object.

DOM Manipulation

Selecting with id

```
document.getElementById("myId")
```

Selecting with class

```
document.getElementsByClassName("myClass")
```

Selecting with tag

```
document.getElementsByTagName("p")
```

DOM Manipulation

Query Selector

```
document.querySelector("#myId / .myClass / tag")  
//returns first element
```

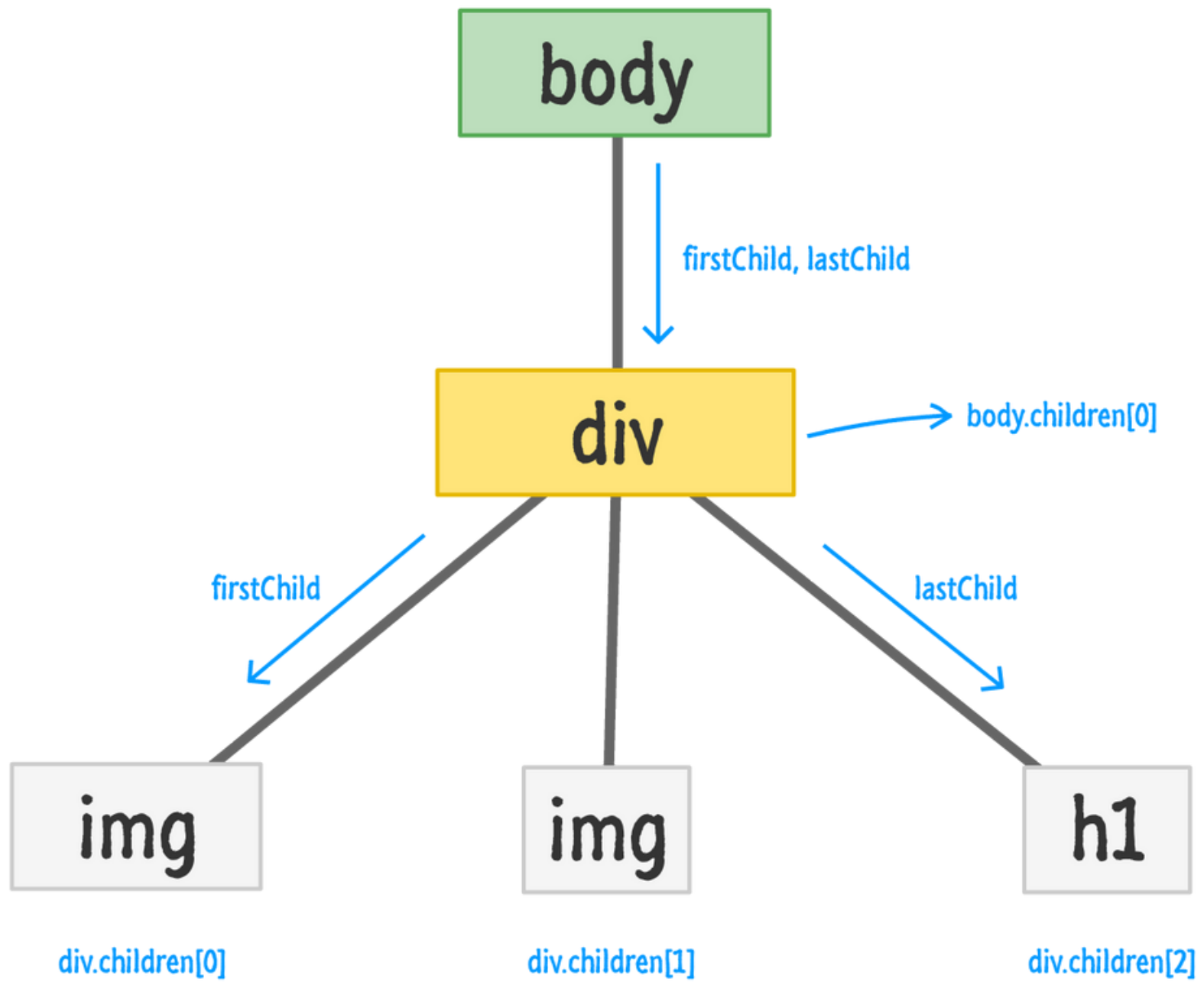
```
document.querySelectorAll("#myId / .myClass / tag")  
//returns a NodeList
```

DOM Manipulation

Properties

- **tagName** : returns tag for element nodes
- **innerText** : returns the text content of the element and all its children
- **innerHTML** : returns the plain text or HTML contents in the element
- **textContent** : returns textual content even for hidden elements

Homework



Let's Practice

Qs. Create a H2 heading element with text - “Hello JavaScript”. Append “from Apna College students” to this text using JS.

Qs. Create 3 divs with common class name - “box”. Access them & add some unique text to each of them.

DOM Manipulation

Attributes

- `getAttribute( attr )` //to get the attribute value
- `setAttribute( attr, value )` //to set the attribute value

Style

- `node.style`

DOM Manipulation

Insert Elements

let el = document.createElement("div")

- `node.append( el )` //adds at the end of node (inside)
- `node.prepend( el )` //adds at the start of node (inside)
- `node.before( el )` //adds before the node (outside)
- `node.after( el )` //adds after the node (outside)

Delete Element

- `node.remove( )` //removes the node

Let's Practice

Qs. Create a new button element. Give it a text “click me”, background color of red & text color of white.

Insert the button as the first element inside the body tag.

Qs. Create a `<p>` tag in html, give it a class & some styling.

Now create a new class in CSS and try to append this class to the `<p>` element.

**Did you notice, how you overwrite the class name when you add a new one?
Solve this problem using classList.**